



Emerging Threats of Harmful Algal Blooms to Seagrass Blue Carbon Resources: Mechanism, Ecological Interactions, and Adaptive Management Strategies

Zhangqi Zuo^{1,2} · Lei Chen¹ · Yingjie Zhu^{1,2} · Yuzhou Huang^{3,4} · Fei Li² · Xi Xiao¹

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Abstract

Purpose of Review Harmful algal blooms (HABs) present a growing threat to seagrass ecosystems, significantly impacting their ecological functions and blue carbon potential. Understanding the complex interactions between HABs and seagrasses is crucial for developing adaptive management strategies to protect seagrass ecosystems.

Recent Findings Recent studies reveal that global HAB events have significantly expanded both geographically and in frequency over the past two decades. The geomorphological processes and depositional environments of seagrass meadows, along with the effects of climate change, act as contemporary drivers that influence algal invasion, presence, and retention within seagrass ecosystems. Emerging research demonstrates that macroalgal blooms can significantly accelerate seagrass carbon loss by enhancing decomposition rates and increasing greenhouse gas emissions from blue carbon stocks. Seagrass allelopathy and associated algicidal bacteria play crucial roles in natural HAB control. Advanced monitoring techniques combining artificial intelligence with remote sensing have achieved significant improvements in detecting and tracking HAB events and seagrass ecosystems.

Summary This review provides a comprehensive analysis of HAB-seagrass interactions, documenting diverse types of HABs affecting seagrass beds, including macroalgal and microalgal blooms. We examine key environmental factors contributing to HABs in seagrass ecosystems, particularly eutrophication, global warming, and ocean acidification, and analyze their complex impact mechanisms, including light limitation, resource competition, biogeochemical alterations, and toxin effects. Natural defense mechanisms of seagrasses, particularly allelopathy and algicidal bacteria, offer potential solutions for HAB control. Effective protection of these valuable blue carbon resources requires integrated adaptive management strategies, combining advanced monitoring technologies, water quality improvement measures, and community-based conservation approaches.

Keywords Seagrass · Harm algal blooms · Eutrophication · Climate change · Seagrass conservation

Zhangqi Zuo, Lei Chen and Yingjie Zhu contributed equally to this work.

✉ Xi Xiao
prana@zju.edu.cn

¹ Ocean College, Zhejiang University, 1 Zheda Road, Zhoushan, Zhejiang 316021, China

² College of Agriculture and Biotechnology, Zhejiang University, Hangzhou 310058, China

³ Key Laboratory of Tropical Marine Ecosystem and Bioresource, Fourth Institute of Oceanography, Ministry of Natural Resources, Beihai 536000, China

⁴ Guangxi Key Laboratory of Beibu Gulf Marine Resources, Environment and Sustainable Development, Fourth Institute of Oceanography, Ministry of Natural Resources, Beihai 536000, China

Introduction

Seagrasses are globally distributed marine flowering plants belonging to the monocotyledonous order Alismatales, comprising approximately 70 species across 13 genera in 6 families [1]. These aquatic plants form extensive underwater meadows spanning protected coastal areas of 163 countries along temperate and tropical coastlines worldwide, covering a total area of approximately 160,387 km² [2]. Seagrass ecosystems are found on all continents except Antarctica [3], thriving in shallow marine environments, estuaries, and lagoons, where they occupy soft-bottom sediments from the intertidal zone to depths of up to 90 m [4–6]. As ecological engineers, seagrasses actively shape their marine environment by attenuating wave energy, enhancing water clarity,

and regulating nutrient cycling, thereby creating optimal conditions for their growth and survival [4, 7]. These meadows serve as critical habitats and nursery grounds for diverse marine organisms, including fish and shellfish species, as well as threatened megafauna such as sirenians, sea turtles, and seahorses [5]. Furthermore, seagrass ecosystems play a crucial role in global carbon cycling through their remarkable capacity to sequester organic carbon in underlying sediments. As one of the most efficient blue carbon sinks, seagrass meadows harbor substantial soil carbon stocks, estimated to range from 1732 to 21,000 Tg C [8–10]. The diverse ecological services provided by seagrass ecosystems position them among the most valuable marine ecosystems globally [6].

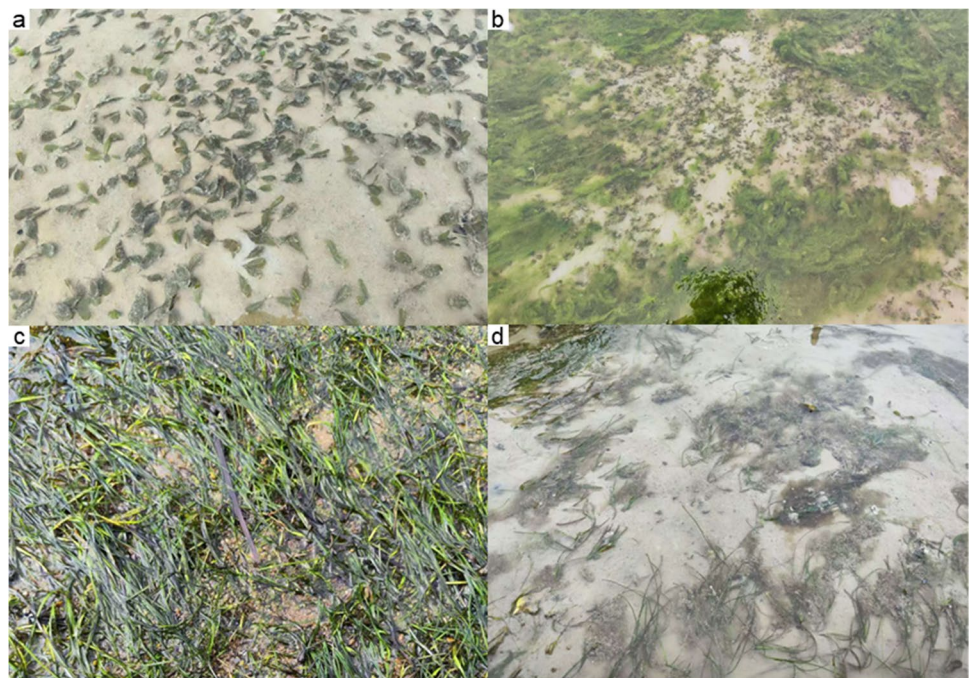
Despite their crucial ecological roles, seagrass meadows are experiencing significant global declines, with an estimated annual loss rate of 0.4–2.6% [11]. Since 1880, approximately 19.1% (5602 km²) of the 29,000 km² surveyed seagrass area has been lost [12]. Biodiversity is also under threat, with 14% of all seagrass species facing elevated extinction risk according to the International Union for Conservation of Nature (IUCN), including three species classified as Endangered on the IUCN Red List: *Phyllospadix japonicus*, *Zostera chilensis*, and *Zostera geojeensis* [13]. The widespread and accelerating losses of seagrass meadows have significant implications for carbon storage, with their degradation potentially releasing up to 299 Tg carbon annually [14]. These declines are driven by multiple threats, including global climate change, such as rising sea surface temperatures, increasing sea levels, and more frequent and intense storms [15–17]. Regional changes in water quality,

along with localized impacts like increased sediment, contaminants, and nutrient loading from human activities, have also negatively affected the health of seagrass ecosystems [18, 19].

These environmental stressors create conditions that favor the proliferation of harmful algal blooms (HABs). HABs are characterized by the rapid growth and accumulation of microscopic algae or phytoplankton, including certain cyanobacteria (blue-green algae), and macroalgae (seaweeds) [20]. In recent years, global climate change and intensified anthropogenic activities have contributed to increasing HAB frequency worldwide. Between 2003 and 2020, algal bloom events were documented in over 80% of coastal nations worldwide, with their geographical coverage expanding by 13.2% alongside a 59.1% increase in occurrence frequency [21]. While the coexistence of algae and seagrass is natural in healthy marine ecosystems, excessive algal growth disrupts the ecological balance of seagrass habitats, resulting in cascading negative impacts [19] (Fig. 1). This degradation jeopardizes the critical ecosystem services that seagrass meadows provide, particularly blue carbon storage, habitat provision for marine species, and coastal protection [22–24]. Understanding the complex interactions between HABs and seagrass ecosystems is therefore essential for safeguarding these valuable habitats and ensuring their continued role in supporting coastal resilience and mitigating climate change.

This review aims to summarize the HABs events in seagrass ecosystems, exploring their various effects, the interactions between HABs and seagrasses, and potential management solutions. By identifying the main factors causing

Fig. 1 Seagrass beds associated with macroalgal blooms in Guangxi, China (**a**, **c** *Halophila ovalis* and *Zostera japonica* during periods without blooms; **b**, **d** same seagrass species during macroalgal bloom events, which typically occur in winter and early spring and may interfere with seagrass canopies)



seagrass decline and understanding how HABs contribute to this problem, we can develop practical approaches to protect and restore these important coastal habitats.

Diversity of HABs Affecting Seagrass Beds

HABs events occurred in seagrass ecosystems have been documented across diverse regions (as Tables 1 and 2), including North America, Europe, Asia, Oceania, and Africa, as well as Japan, Mexico, Malaysia, and Portugal (Fig. 2). In these seagrass habitats, HABs exhibit remarkable diversity in terms of their causative organisms, which can be broadly categorized into macroalgal and microalgal blooms.

Macroalgal Blooms

Macroalgal blooms, typically dominated by large, multicellular algae, can form extensive mats or floating aggregations that disrupt the structure and function of seagrass meadows. These blooms appear in various forms within seagrass ecosystems. One of the most common types of macroalgal blooms is the green tide, predominantly caused by genera such as *Caulerpa*, *Chaetomorpha*, *Cladophora*, *Enteromorpha*, *Halimeda*, and *Ulva*, with *Ulva* being the most frequently observed. Green algal blooms can form dense floating mats that accumulate above seagrass meadows, creating layers of varying thickness and coverage that blanket the underlying vegetation and disrupt ecosystem function [68]. A particularly severe green tide event occurred in the Knysna Estuary, South Africa, where seasonal blooms of *Ulva* began in the late austral summer of 2015. These blooms led to extensive seagrass mortality, including in areas of high national conservation and biodiversity importance, ultimately leaving behind bare muddy sediment and disrupting essential habitat for marine life [38]. Given that the Knysna Estuary provides vital ecosystem services—supporting fish nurseries valued at approximately \$14 million annually, nutrient cycling services worth \$4.5 million per year, and tourism revenue estimated at \$85 million annually—the degradation of seagrass meadows likely resulted in substantial economic losses [38].

Red algae blooms featuring genera such as *Botryocladia*, *Gracilaria*, and *Gracilariopsis* constitute another significant form of macroalgal proliferation. These branched red algal species typically grow in dense clusters among the basal portions of seagrass shoots, sometimes inhibiting seagrass growth [68]. For example, *Gracilaria vermiculophylla* has been shown to negatively affect the above-ground biomass of *Zostera marina* [69]. A particularly severe outbreak of *G. vermiculophylla* has been observed in areas where seagrass

meadows have already declined, resulting in the formation of dense mats that reduce light penetration, induce hypoxic conditions, and physically smother native seagrass species [70]. In August 2005, a harmful algal bloom of *Botryocladia occidentalis* occurred in the coastal waters of Lee County, FL, USA, closely associated with the enrichment of land-derived nutrients [44]. The bloom led to the formation of dense algal mats, which accumulated in large quantities along the coastline, making the beach unsuitable for recreational activities [44]. Additionally, the algal mats attached to the feeding tubes of polychaete *Chaetopterus variopedatus* in seagrass beds, disrupting local benthic communities and further compromising the ecosystem, with significant negative impacts on both the tourism industry and local water resources [44]. Additionally, the cost of beach cleanup efforts in Lee County rose sharply, increasing from minimal amounts to over \$260,000 during the 2003/2004 fiscal year as a result of the rhodophyte blooms [44].

Golden tides, often attributed to *Sargassum* species, are characterized by their distinctive golden-brown color and extensive floating mats. Between mid-2014 and the end of 2015, the Mexican Caribbean coast experienced an unprecedented influx of drifting *Sargassum* spp., which accumulated along the shorelines, leading to the mortality of nearshore seagrasses, associated fauna, and corals [49]. Recovery of the affected seagrass meadows may take years or even decades. In addition, the invasion of *Sargassum muticum* in *Z. marina* seagrass beds, particularly within soft-sediment habitats, poses a significant threat to these ecosystems. *S. muticum* is capable of colonizing these areas by exploiting the dense rhizome matrix of *Z. marina*, which traps drifting fragments, allowing the algae to anchor in previously unsuitable sediments [71]. Since 2015, large-scale blooms of *Sargassum natans* and *Sargassum fluitans* have extensively covered the Caribbean coastline of Mexico, affecting approximately 400,000 hectares of *T. testudinum* seagrass beds [72, 73]. Over the past few decades, the quantity of massive macroalgal mats, which accumulate and wash ashore along the coastline, has steadily increased. These mats reach the shallow bays of Mexico, where they remain for several months each year. The persistent presence of *Sargassum* spp. has significantly threatened the health of local seagrass and coral reef ecosystems, and the blooms have disrupted the region's coastal tourism industry, which generates approximately \$10 billion annually [73, 74].

Microalgal Blooms

Microalgal blooms in seagrass ecosystems are characterized by diverse microscopic algae species, including dinoflagellates, raphidophytes, pelagophytes, and cyanobacteria. These organisms can exist as both planktonic and epiphytic forms,

Table 1 List of typical macroalgal blooms occurring in seagrass habitats

Macroalgae bloom species	Seagrass species	Effect	Outbreak time	Location	Reference
Class	Species				
Ulvothyceae	<i>Caulerpa prolifera</i>	Did not significantly reduce seagrass biomass or distribution	Early summer (July) of 2014	Ria Formosa Natural Park, Algarve, Portugal	[25]
	<i>Caulerpa prolifera</i>	Reduced seagrass biomass and shoot density by up to 50%	April–October 2000	Indian River Lagoon, FL, USA	[26]
	<i>Chaetomorpha linum</i>	Reduced seagrass biomass by up to 65%, decreased shoot density by 55%	From 5 September to 8 December 2019	Bekalta, Eastern Coast of Tunisia	[27]
	<i>Chaetomorpha ligustica</i>	Reduced seagrass coverage by 50%	From 28 October to 12 December 2019	Pāuatāhau Inlet, Wellington Region, New Zealand	[28]
	<i>Chaetomorpha linum</i>	Reduced seagrass biomass, causing a 50% decline in seagrass health	From February to May 2013, and again from September to October of the same year	San Andrés Island, Southwestern Caribbean, Colombia	[29]
	<i>Cladophora vagabunda</i>	Reduced seagrass biomass by up to 50%	June 2018	Yellow River Estuary, Shandong Province, China	[30]
	<i>Enteromorpha intestinalis</i>	Reduced seagrass by over 50%	Mid-September 2002	Tuggerah Lakes Estuary, New South Wales, Australia	[31]
	<i>Enteromorpha radiata</i>	Reduced seagrass coverage to zero	September 1991	Hayling Island, Hampshire, UK	[32]
	<i>Halimeda incrassata</i>	Reduced seagrass growth by 10.3% in shoot size and 33.3% in biomass	From July through November 1997	John Pennekamp State Park, FL, USA	[33]
	<i>Ulva intestinalis</i>	Reduced seagrass biomass	From May to September 2018	Yellow River Estuary, Shandong Province, China	[34]
	<i>Ulva lactuca</i>	Reduced soil organic carbon stocks in seagrass meadows by 37%	January 2018	Xincun Bay Lagoon, Hainan Island, China	[35]
	<i>Ulva pertusa</i>	Reduced seagrass belowground biomass by 53% and altered carbon/nitrogen ratios	June 2012	Swan Lake, Shandong Province, China	[36]
	<i>Ulva reticulata</i> Forsskål	Increased seagrass leaf size but reduced seagrass health	July 2014	Merambong Shoal, Sungai Pulai Estuary, Johor, Malaysia	[37]
	<i>Ulva</i> sp.	Reduced seagrass coverage by over 80%	In the late austral summer of 2015	Knysna Estuarine Bay, Western Cape, South Africa	[38]
	<i>Ulva</i> sp.	Reduced seagrass shoot density, seedling recruitment, and growth rate	August and November 2000	Iwakuni, Seto Inland Sea, Japan	[39]
	<i>Ulva</i> sp.	Reduced seagrass shoot density by up to 83%	April 2009	Coos Bay, OR, USA	[40]

Table 1 (continued)

Macroalgae bloom species	Seagrass species	Effect	Outbreak time	Location	Reference
Class	Species				
Florideophyceae	<i>Ulva</i> sp.	Reduced seagrass shoot density by 45%, biomass by 53%, and carbon storage by 82%	From 27 July to 8 August 2007	Prince Edward Island, Canada	[41]
	<i>Ulva</i> sp.	Reduced seagrass seed bank density (5–35 seeds/m ²), leading to population extinction	August and October 2004	Wadden Sea	[42]
	<i>Ulvaria obscura</i>	Reduced seagrass shoot density by up to 54%	12 July 1999	Blakely Island, WA, USA	[43]
	<i>Botryocladia occidentalis</i>	Reduced seagrass biomass and area, decreased biodiversity	8–10 August (dry season) and 28–29 October (wet season), 2004	Lee County, FL, USA	[44]
	<i>Gracilaria comosa</i>	Reduced seagrass biomass, growth, and survival by up to 85%	2011	Swan River Estuary, Perth, Western Australia	[45]
	<i>Gracilaria</i> sp.	Reduced seagrass shoot density by up to 100%, decreased growth rates by 55%, and inhibited recruitment	Mid-April 2000	Waquoit Bay, MA, USA	[46]
	<i>Gracilaria vermiculophylla</i>	Reduced seagrass photosynthesis by 25%, decreased survival from 74% to 21%	In August, October and November 2007	Funen Island, Denmark	[47]
	<i>Gracilariaopsis</i> sp.	Reduced seagrass shoot density by 49%, biomass, and growth rates	June 2005	Tomales Bay, CA, USA	[48]
Phaeophyceae	<i>Thalassia testudinum</i>	Reduced seagrass belowground biomass by 99.5%, causing a dramatic reduction in seagrass area	From mid-2014 until the end of 2015	Mirador Nizuc, Quintana Roo, Mexico	[49]
Unknown*	<i>Thalassia testudinum</i>	Reduced seagrass above-ground biomass by 25%	September 1997 (lasting one year)	Biscayne Bay, FL, USA	[50]
	<i>Thalassia testudinum</i>	Decreased seagrass biomass and photosynthetic efficiency	Autumn 2003	Port Aransas, TX, USA	[51]
	<i>Zostera marina</i>	Reduced seagrass biomass, shoot density, and reproductive success with up to 67%	Mid-April 2000	Waquoit Bay, MA, USA	[46]
	<i>Zostera marina</i>	Reduced seagrass area and biomass of 80–96%	February 2008	Waquoit Bay, MA, USA	[52]

*Unknown refers to cases where the specific algal species were not mentioned in the cited literature

Table 2 List of microalgal blooms on seagrass

Microalgae bloom species	Seagrass species	Effect	Outbreak time	Location	Reference
<i>Heterosigma akashiwo</i>	<i>Zostera marina</i>	Reduced seagrass shoots density, biomass, and biodiversity	In late May 2002	Jindong Bay, Korean Peninsula, South Korea	[53]
<i>Phaeophyceae</i> sp.	<i>Halodule wrightii</i>	Reduced seagrass biomass, distribution (over 9 km ²), and ecosystem biodiversity	In June 1990, continuously through December 1995	Laguna Madre, TX, USA	[54]
<i>Prorocentrum concavum</i>	<i>Halassia hemprichii</i> and <i>Enhalus acoroides</i>	Reduced seagrass area and biodiversity	August 2018	Xincun Bay Lagoon, Hainan Island, China	[55]
<i>Aureococcus anophagefferens</i>	<i>Zostera marina</i>	Reduced seagrass distribution and biomass with up to 4.5 km ²	The past 4 summers (1985–1988)	Long Island, NY, USA	[56]
<i>Aureoumbra lagunensis</i>	<i>Halodule wrightii</i> , <i>Syringodium filiforme</i> and <i>Thalassia testudinum</i>	Caused up to 95% seagrass loss, and reduced biomass, exacerbating ecosystem degradation.	Between 2011 and 2017	Indian River Lagoon, FL, USA	[57]
<i>Prorocentrum lima</i> , <i>Mastogloia corsicana</i> and <i>Actinocyclus normanii</i>	<i>Thalassia testudinum</i>	Reduced seagrass health, posed threats to ecosystem biodiversity	2015	Barú Island, Bolívar, Colombia	[58]
<i>Lyngbya majuscula</i>	<i>Halodule wrightii</i>	Reduced seagrass biomass and density, increased leaf length, altered invertebrate populations, and contributed to long-term ecosystem degradation	During the summer of 2006	Indian River Lagoon, FL, USA	
<i>Nitzschia</i> sp., <i>Thalassiosira</i> sp., <i>Chaetoceros</i> sp., <i>Flagellaria</i> sp., <i>Thalassiothrix</i> sp., <i>Melosira</i> sp., <i>Oscillatoria</i> sp., <i>Spirogyra</i> sp. and <i>Dinophysis</i> sp.	<i>Halophila ovalis</i> , <i>Enhalus acoroides</i> and <i>Thalassia hemprichii</i>	Reduced seagrass biomass and biodiversity	2015	Pegametan Bay, Bali, Indonesia	[59]
<i>Aureoumbra lagunensis</i>	<i>Halodule wrightii</i>	Reduced seagrass biomass	During the second half of 2012	Mosquito Lagoon and Indian River Lagoon, FL, USA	[60]
<i>Karenia brevis</i>	<i>H. wrightii</i> and <i>S. filiforme</i> <i>T. testudinum</i>	Impacted seagrass communities	In September 2007	East Coast of Florida, USA	[61]
<i>Lyngbya majuscula</i> and <i>Karenia</i> spp.	<i>Halodule wrightii</i>	Reduced seagrass area and biomass while altering ecosystem biodiversity	During July 2006	Sanibel and Captiva Islands, FL, USA	[62]
<i>Lyngbya majuscula</i>	<i>Halophila ovalis</i> , <i>Halophila spinulosa</i> , <i>Syringodium isoetifolium</i> , <i>Zostera capricorni</i> , <i>Halodule uninervis</i> and <i>Cymodocea serrulata</i>	Reduced seagrass area (80 km ²) and biomass, leading to biodiversity decline	In 2000 and 2001	Moreton Bay, Queensland, Australia	[63]
<i>Lyngbya majuscula</i>	<i>Zostera capricorni</i> , <i>Halophila spinulosa</i> , <i>H. ovalis</i> and <i>Halodule uninervis</i>	Reduced seagrass biomass by up to 76%, with significant declines in leaf and root biomass	Over the period January–March 2000	Deception Bay, Queensland, Australia	[64]
<i>Prorocentrum lima</i>	<i>Zostera</i> spp. and <i>Ruppia</i> spp.	Increased seagrass colonization	During summer 2002	Fleet Lagoon, England, UK	[65]

Table 2 (continued)

Microalgae bloom species	Seagrass species	Effect	Outbreak time	Location	Reference
<i>Aureocoumbra lagunensis</i>	<i>Thalassia testudinum</i>	Reduced seagrass biomass by up to 75%	2004	Lower Laguna Madre, South Texas, USA	[66]
<i>Microcoleus lyngbyaceus</i>	<i>Thalassia testudinum</i>	Reduced seagrass cover from 52% to 9%	From September 1998 to February 2000	Inshore Waters of Puerto Rico, Caribbean Sea	[67]

with some species producing harmful toxins that affect marine life.

Red tides, primarily caused by dinoflagellates and raphidophytes, represent a significant category of microalgal blooms. *Heterosigma akashiwo*, a raphidophyte species, has caused devastating red tide events, such as the severe bloom along the southern coast of Korea in May 2002 that resulted in the near-complete loss of *Z. marina* populations [53]. Dinoflagellates like *Prorocentrum* species can occur both as planktonic and epiphytic forms. A notable example is the benthic bloom of *Prorocentrum concavum* observed in Xincun Bay, Hainan Island, which demonstrated toxicity to brine shrimp larvae (*Artemia salina*), highlighting the potential threat posed by toxic benthic dinoflagellate blooms to benthic marine ecosystems centered around seagrass beds, as well as to public health [55].

Brown tides, formed by pelagophytes including *Aureococcus anophagefferens* and *Aureocoumbra lagunensis*, produce unique brown-colored water in coastal bays. Blooms of *Aureococcus*, a non-motile spherical unicellular alga, have been documented in various regions [75]. During 1985 to 1987, massive *Aureococcus* blooms occurred in the bays of eastern and southern Long Island, with cell densities exceeding ($>10^6$ cells/ml) [76, 77]. These intense blooms devastated local *Z. marina* meadows, which serve as essential habitats for diverse marine organisms, including larval fish and bay scallops. The ecological damage extended to the commercial sector, as the decimation of bay scallop (*Argopecten irradians*) populations, through both direct mortality and recruitment failure, led to the collapse of eastern Long Island's lucrative scallop fishery [78].

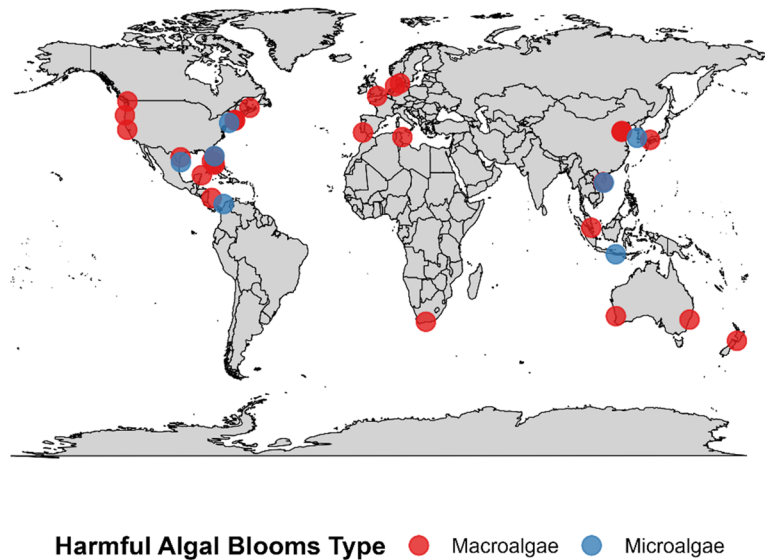
Cyanobacterial blooms, particularly those formed by species such as *Lyngbya majuscula* and *Moorea producens*, create dense benthic mats on sediments and seagrass blades, leading to significant ecological consequences. In regions like the Indian River Lagoon, Florida, *L. majuscula* blooms have been shown to drastically reduce the density and biomass of *Halodule wrightii* [79]. The *L. majuscula* bloom in the waters of Moreton Bay, located along the southeastern coast of Queensland, Australia, impacts an area that supports extensive *Zostera ramricorni* seagrass beds, mangroves, and coral communities [80]. During the bloom, *L. majuscula* loosely attaches to seagrass, macroalgae, and sediment or rocky substrates [81]. The bloom leads to smothering of seagrass and mangrove seedlings, resulting in oxygen depletion and disruption of local benthic communities [81].

Environmental Factors Contributing to HABs in Seagrass Ecosystems

Seagrass meadows are susceptible to both macroalgae and microalgae blooms, as they provide an ideal substrate and a source of nutrients that support the growth and

Fig. 2 Global distribution of harmful algal blooms in seagrass beds

Global Distribution of Harmful Algal Blooms in Seagrass Beds



reproduction of algae [82–84]. Additionally, the structure and characteristics of seagrass blades make them more likely to capture floating macroalgae [85, 86], further increasing the likelihood of HAB outbreaks in seagrass meadows. In recent years, rapid global environmental changes have increased the likelihood of large-scale HABs in seagrass meadows, which can inhibit the growth of seagrass and contribute to the global degradation of seagrass ecosystems [82]. Environmental factors contributing to HABs in seagrass ecosystems are complex and interconnected, often acting synergistically to create conditions that favor excessive algal proliferation while compromising seagrass resilience (Fig. 3).

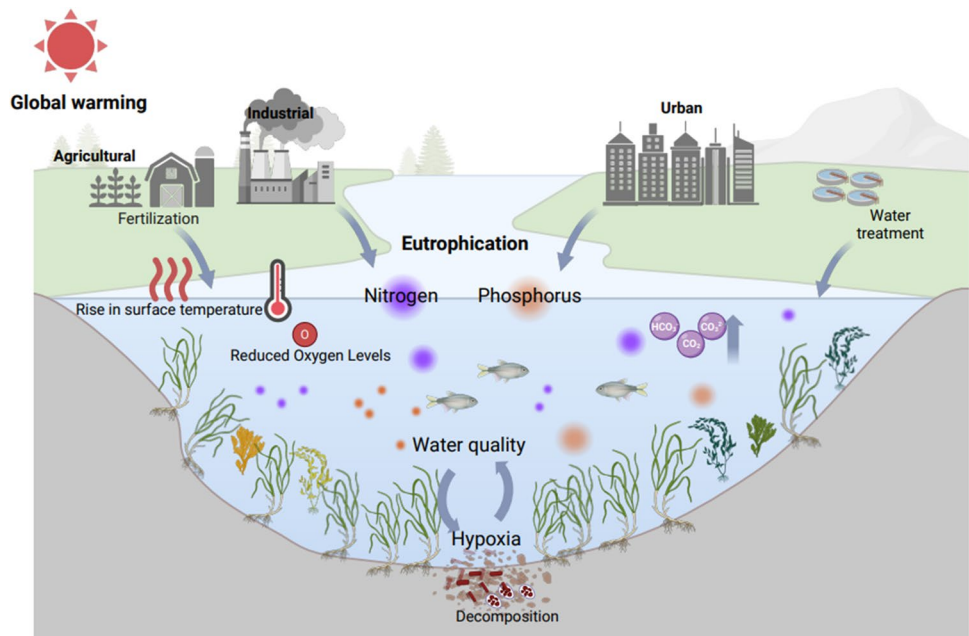
Geomorphology and Sedimentation

Seagrass meadows occur across a range of geomorphological settings, from sheltered embayments to open coastal areas [87]. In relatively low-energy environments, the opportunistic macroalgae such as *Vaucheria* may compete with *Z. noltii* for space and resources, particularly in areas with high sedimentation rates [88]. Field observations from Banc d'Arguin in Mauritania have recorded higher algal biomass in sheltered areas [88]. Similarly, studies in the Yellow River Estuary show that macroalgae may compete with seagrasses in nearshore zones but facilitate their growth further offshore, reflecting a gradient of interaction from land to sea [30]. Drift macroalgae, initially attached to seagrass leaves or other substrates, are detached by currents or wave action and redistribute across meadows [89]. Their redistribution across seagrass beds is influenced by hydrodynamic conditions and structural features of the

meadow. For example, surveys from the northern Gulf of Mexico revealed that macroalgal accumulation is not random but tends to be highest in areas with greater shoot density and intermediate canopy height, where the physical structure of the seagrass canopy modulates both water flow and deposition patterns [89].

In addition to natural geomorphological influences, human activities such as anchoring and dredging can damage seagrass rhizomes and shoots, leading to fragmentation and the creation of vulnerable patches [90]. These disturbed areas provide favorable conditions for invasive species like *Caulerpa racemosa*, which is widespread in the Northwest Mediterranean. This species tends to expand at the edges of degraded seagrass beds, while dense and healthy stands of *Posidonia oceanica* can suppress its vertical and horizontal growth [90]. Fragmented meadows with exposed sandy or matte patches are especially prone to invasion. Moreover, anthropogenic disturbances may modify meadow bathymetry and hydrodynamics, thus promoting macroalgal proliferation and reducing the meadow's carbon storage capacity [91]. Recent research also emphasizes the role of sediment characteristics in regulating invasion success. For example, the growth of *Caulerpa cylindracea* fragments is reduced in sediments from *P. oceanica* habitats compared to those from its own habitat [92]. This difference is likely driven by variations in microbial communities and sediment chemistry. These findings underscore that belowground processes, in addition to canopy integrity, are critical in determining resistance to invasive macrophytes and in shaping the resilience of seagrass ecosystems under increasing anthropogenic stress.

Fig. 3 Environmental factors contributing to HABs in seagrass ecosystem (created in <https://BioRender.com>)



Eutrophication

Eutrophication represents a primary driver of HABs in seagrass ecosystems, triggering a complex cascade of ecological changes. The widespread coastal eutrophication stems primarily from two sources: the discharge of nutrient-rich wastewater through rivers and estuaries from inland watersheds, and the extensive use of agricultural fertilizers [93]. It is estimated that approximately 24% of anthropogenic nitrogen reaches coastal ecosystems after being released into coastal watersheds, contributing to eutrophication [94]. Eutrophication in coastal areas has become a global issue, particularly in the Northern Hemisphere, including the western margins of the Atlantic and Pacific Oceans, as well as European coastal waters [94–96]. Recent coastal surveys in the USA and the European Union reveal that 78% of US coastal waters and 65% of Atlantic coastal waters in Europe exhibit symptoms of eutrophication [94]. During the twentieth century, the anthropogenic input of nitrogen and phosphorus into coastal ecosystems through river discharges increased, becoming a primary cause of eutrophication and subsequent ecosystem degradation in global coastal areas [95, 97]. Elevated nitrogen and phosphorus levels often promote HABs in seagrass meadows, causing both the proliferation of harmful algae and the death of seagrass [93].

This nutrient-enriched environment in seagrass beds creates optimal conditions for algal proliferation, initiating a transition from seagrass-dominated to algae-dominated systems [98]. Under these nutrient-enriched conditions, when nutrient content exceeds critical thresholds, algae will

gradually replace seagrass due to their rapid growth rates, superior nutrient uptake efficiency, and shading effects that reduce light availability for underlying seagrass beds [99]. Both microalgae blooms (including dinoflagellates, cyanobacteria, and diatoms) and blooming macroalgae, particularly opportunistic green algae (*Ulva*, *Cladophora*) and red algae (*Gracilaria*, *Gracilariopsis*), exhibit rapid proliferation in response to the alteration of N:P:S stoichiometric ratios and the removal of nutrient limitations in marine ecosystems [36, 100]. *Gracilaria tikvahiae* McLachlan, a red macroalga, has been observed displacing seagrass beds in the Indian River Lagoon, particularly under conditions of nitrogen enrichment and phosphorus limitation common in urbanized estuaries [44]. Similarly, microalgal blooms can cause catastrophic seagrass losses, as demonstrated by persistent brown tides of *Aureoumbra lagunensis* in the Indian River Lagoon, which resulted in approximately 95% seagrass decline between 2011 and 2017 [57].

Moreover, eutrophication directly harms seagrass growth, as elevated NH_4^+ concentrations impair seagrass development, while excessive nitrate uptake induces carbon limitation in these marine plants [101]. This leads to the decline of seagrass meadows, allowing algae to gain a competitive advantage over seagrass in nutrient acquisition [98]. As seagrass health deteriorates and coverage declines, a positive feedback loop emerges: sediment destabilization accelerates nutrient release into the water column [101], while microbial decomposition of fallen seagrass leaves further contributes to the pool of inorganic nitrogen and phosphates [102], which amplifies eutrophication and triggers the outbreak of HABs.

Global Warming

The massive emissions of greenhouse gases such as water vapor (H_2O), carbon dioxide (CO_2), nitrous oxide (N_2O), methane (CH_4), and ozone (O_3) from human activities have led to global warming. From 1880 to 2012, the global average temperature increased by 0.85°C (within a range of 0.65 – 1.06°C) [103]. Since most of the Earth's surface is covered by seawater, and water has a higher specific heat capacity, the majority of the excess heat has been absorbed by the oceans, leading to a rise in sea surface temperature. In the upper 75 m of the ocean, the average temperature increased at a rate of 0.11°C per decade between 1971 and 2010 [103].

Global warming-induced changes in sea surface temperature significantly influence HAB dynamics in seagrass ecosystems. Elevated surface temperatures intensify water column stratification, creating favorable conditions for algal proliferation by reducing vertical mixing and maintaining algal populations within the photic zone; this thermally induced stratification leads to earlier bloom initiation and extended bloom duration [104]. Furthermore, temperature-driven alterations in oceanic circulation patterns enhance upwelling intensity, particularly along eastern boundary currents, thereby elevating nutrient flux to surface waters and optimizing stoichiometric conditions ($\text{N:P} > 16$) for algal biomass accumulation [21].

Concurrently, temperature elevation within the 5 – 25°C range enhances enzymatic catalytic efficiency in algal cells, accelerating photosynthetic processes and shortening cellular division cycles [21]. For instance, diatom proliferation rates increase by 33% when temperatures rise from 15°C to 20°C , significantly compressing their reproductive timelines [21]. Temperature-mediated enhancement of metabolic processes and growth rates is particularly pronounced in certain seagrass HAB species, notably *Prorocentrum lima*. Under moderate warming (15 – 25°C), accelerated metabolic rates in *P. lima* not only elevate growth (peaking at 0.18 day^{-1}) but also activate toxin synthesis pathways, increasing cellular okadaic acid production to 6.04 pg cell^{-1} [105]. On Isla de Barú in the Caribbean, peak *P. lima* densities of $52 \pm 48\text{ cells/g}$ wet weight substrate have been recorded when temperatures reach 25°C [58]. Furthermore, even extreme temperature (30°C) stress-induced metabolic shifts drive persistent toxin accumulation [105]. The dual pressures of proliferating cell densities and intensified toxicity enable *P. lima*-dominated HABs to outcompete seagrass-associated microbiota, accelerating seagrass degradation. Meantime, global warming has a direct negative effect on the growth of seagrass. Heat stress reduces seagrass resilience by decreasing photosynthetic efficiency, increasing respiratory oxygen consumption, and causing oxidative damage [106]. These physiological declines can lead to seagrass die-offs, the

release of stored nutrients, and sediment destabilization, thereby creating favorable conditions for the development of algal blooms.

Ocean Acidification

Atmospheric carbon dioxide (CO_2) increased from 278 ppm to 400 ppm during the industrial period; the ocean acts as a climate integrator, capturing 28% of anthropogenic CO_2 emissions since 1750 [107], as well as CO_2 emissions from underwater volcanic vents [108]. This increase in CO_2 leads to a decrease in seawater pH, reducing the saturation of calcium carbonate minerals and triggering ocean acidification.

Seagrasses are typically limited by carbon availability. Ocean acidification modifies the chemical environment of seagrass meadows, leading to elevated availability of photosynthetic substrates (carbon dioxide and bicarbonate) [108, 109]. While such conditions enhance the photosynthetic efficiency and productivity of seagrasses [110, 111], the ecological consequences are far more complex. A field study of volcanic carbon dioxide vents demonstrated that seagrass productivity peaked at a mean pH of 7.6 as pH decreased from normal levels (8.1 – 8.2) to lower values (mean 7.8 – 7.9 , minimum 7.4 – 7.5) [112]. However, communities showed marked composition shifts with calcareous organisms including scleractinian corals, sea urchins, and coralline algae experiencing substantial reductions [112]. Furthermore, these potential productivity benefits from CO_2 -enriched seawater may be counterbalanced by the negative effects of rising temperatures on seagrass vegetative growth [110, 113].

Ocean acidification not only affects seagrasses directly but also drives changes in epiphytic and planktonic microalgal communities associated with seagrass ecosystems. As pH decreases, the increase in HCO_3^- and carbonate CO_3^{2-} concentrations makes the deposition and stabilization of calcium carbonate more difficult [108]. Since calcium carbonate is an essential component of the shells or skeletons of many epiphytic algae, particularly calcareous algae [114], the reduction of calcium carbonate limits their ability to survive in seagrass ecosystems [108]. In contrast, non-calcifying microalgae, such as diatoms, are favored under acidified conditions. The increase in CO_2 concentrations in surface waters under ocean acidification leads to a down-regulation of the diatom carbon concentrating mechanisms (CCMs), saving energy that can be redirected for growth or other metabolic processes [115]. For example, *Chaetoceros muelleri*, a bloom-forming non-calcifying diatom, exhibits significant increases in CO_2 uptake, with their carbon fixation rate rising by approximately 20% in acidified seawater [115]. This shift favors the proliferation of opportunistic, fast-growing taxa and elevates the risk of HAB events, especially in nutrient-rich seagrass environments. Such

trends have already been observed in acidified *P. oceanica* meadows, where non-calcifying algae thrive due to diminished competition from calcifying species (e.g., *Myrionema orbiculare*) and synergistic effects between reduced pH and light availability [109].

Impact Mechanisms of HABs on Seagrass Ecosystems

Light Limitation

Seagrasses fundamentally depend on adequate light penetration for survival and growth [116]. The photosynthetic process in seagrasses requires light to successfully navigate multiple barriers—traversing the water column, penetrating the leaf canopy, passing through potential epiphytic layers, and finally reaching the photosynthetic machinery within the leaf tissue [117]. Each stage introduces light attenuation, making seagrasses vulnerable to light-reducing factors [118].

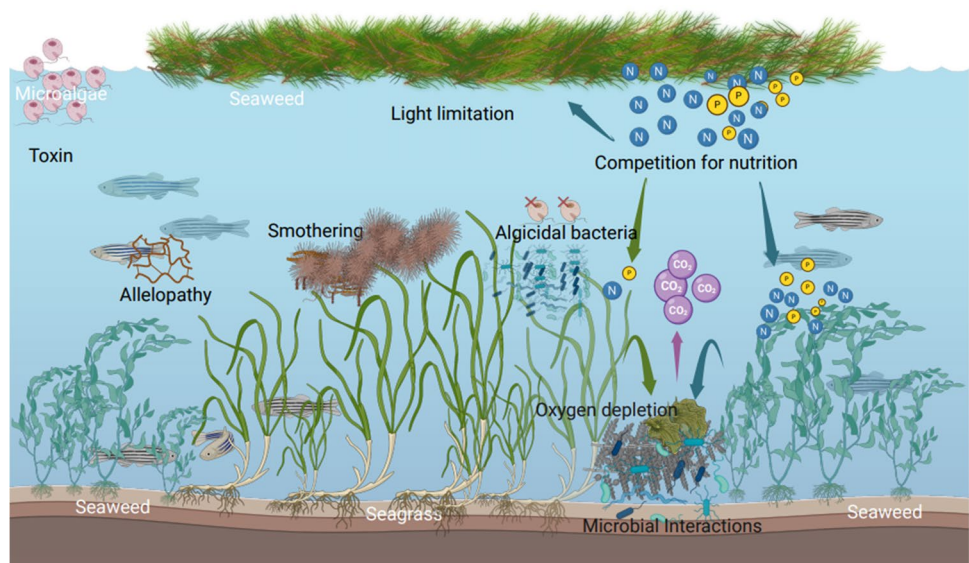
Opportunistic macroalgal blooms, particularly *Ulva* green tides and *Sargassum* golden tides, create substantial challenges for seagrass communities by forming dense canopies that significantly diminish light availability [38, 49, 119] (Fig. 4). In extreme cases, *Ulva* in Hiroshima Bay have been reported to block over 90% of incident light, leading to severe growth inhibition, recruitment failure, and large-scale loss of seagrass meadows [39]. Beyond the quantitative reduction in light availability, these macroalgal blooms also induce qualitative shifts in the spectral composition reaching seagrass tissues. *Ulva thalli* preferentially absorb wavelengths in the blue and red portions of the spectrum, resulting in an enrichment of green

wavelengths in the remaining light that penetrates to the seagrass canopy [120]. Experimental studies on *Z. marina* have demonstrated that while seagrasses can still utilize green light, the presence of *Ulva*-induced pH shifts—often reaching values as high as 10—can significantly reduce photosynthetic efficiency by inhibiting electron transport rates by up to 75% [120].

Dense microalgal blooms substantially alter water color and transparency, creating persistent light limitation [121]. Brown tides, primarily composed of pelagophytes including *Aureococcus anophagefferens* and *A. lagunensis*, have emerged as particularly problematic algal assemblages across shallow estuaries in the USA and South Africa over recent decades [77]. These microalgae exhibit optimal growth under low irradiance conditions, allowing them to proliferate in turbid environments [77]. In Laguna Madre, these persistent blooms reduced water column light penetration by 31–63% annually, pushing underwater irradiance below critical physiological thresholds and triggering widespread seagrass decline [54, 122].

Seagrass leaf surfaces frequently host complex microbial communities, which form dense biofilms consisting of bacteria, diatoms, and filamentous algae that substantially reduce light availability by obstructing the transmission of photosynthetically active radiation [123]. Experimental measurements have demonstrated that thick epiphytic biofilms can attenuate more than 50% of incident photosynthetically active radiation (PAR), diminishing both the quantity and spectral quality of light reaching photosynthetic tissues [124, 125]. Particularly detrimental are cyanobacterial mats directly covering seagrass leaves, as observed in *H. wrightii* meadows within the Indian River Lagoon, creating an additional photosynthetic barrier and contributing to the loss of *H. wrightii* biomass [79].

Fig. 4 Interactions between HABs and seagrass ecosystems (created in <https://BioRender.com>)



Resource Competition

Various algal species, including red drift rhodophytes (*Gracilaria* and *Hypnea*) [126], floating green macroalgae (*Ulva*) [127], rhizophytic green algae (*Caulerpa*) [26, 128], and epiphytic red algae (*Smithora*) [129], engage in competitive interactions with seagrasses for resources such as light and nutrients, particularly in eutrophic coastal environments (Fig. 4). Morphological characteristics strongly influence competitive outcomes. Sheet-like forms of macroalgae like *U. lactuca* possess high surface area-to-volume ratios ($565.2 \pm 2.15 \text{ cm}^2 \text{ g dry wt}^{-1}$), substantially higher than finely branched *Hypnea musciformis* ($110.7 \pm 3.97 \text{ cm}^2 \text{ g dry wt}^{-1}$) and coarsely branched *G. tikvahiae* ($91.1 \pm 1.81 \text{ cm}^2 \text{ g dry wt}^{-1}$), enabling *U. lactuca* to achieve rapid biomass doubling times of just two days under favorable conditions [126].

Differential nutrient acquisition strategies further shape competitive dynamics between seagrasses and algae. Seagrasses typically thrive in oligotrophic conditions with sporadic nitrogen availability, and their competitive edge lies in their ability to efficiently take up nitrogen during nutrient pulses [130]. *Ulva* readily assimilates different forms of dissolved inorganic and organic nitrogen in the sequence of NO_3^- (nitrate) > NH_4^+ (ammonium) > $\text{CO}(\text{NH}_2)_2$ (urea) > $\text{C}_2\text{H}_5\text{NO}_2$ (glycine) and $\text{C}_3\text{H}_7\text{NO}_2$ (alanine), allowing dominance in nitrogen-enriched waters [131]. Rhodophytes employ an alternative strategy, preferentially absorbing ammonium while exhibiting enhanced adaptation to nitrogen-limited conditions through broader environmental tolerance ranges and specialized nitrogen storage in accessory pigment complexes [132]. Light availability significantly affects nutrient uptake dynamics. *Ulva pertusa* showed substantially higher absorption rates for ammonium and nitrate under high-light conditions (14.67 ± 2.50 and $1.29 \pm 0.16 \text{ mg}^{-1} \text{ DW h}^{-1}$) compared to low-light conditions (4.52 ± 0.95 and $0.18 \pm 0.12 \text{ mg}^{-1} \text{ DW h}^{-1}$), demonstrating the interplay between light and nutrient competition [127].

Rhizophytic green algae use rhizoid networks to anchor in unconsolidated sediments, often forming an understory in seagrass beds or establishing mixed or monospecific stands [33]. While they can initially facilitate seagrass colonization by stabilizing sediments and providing organic matter during ecosystem development, rapid expansion of species like *C. prolifera* creates competitive pressure [133]. Studies in the Ria Formosa lagoon demonstrated that *C. prolifera* typically outpaces seagrass *Cymodocea nodosa* in acquiring both ammonium and amino acids through above- and belowground structures [133].

Biogeochemical Alterations

HABs significantly alter biogeochemical processes within seagrass ecosystems through multiple mechanisms. Red tide events, particularly those dominated by *Heterosigma akashiwo*, release mucilaginous exudates that coat seagrass surfaces, creating physical barriers to metabolite exchange and intensifying the smothering effect beyond simple shading, as documented in Korean coastal waters where *Z. marina* populations experienced precipitous decline [53]. Dense macroalgal mats physically reduce water circulation, diminishing oxygen and nutrient transport throughout the water column [98, 134]. Rapidly growing macroalgae absorb nutrients and fix carbon, while also releasing organic compounds such as dimethylsulfoniopropionate (DMSP), dopamine, and producing reactive oxygen species (ROS) that modify surrounding water chemistry and affect other organisms [135]. As macroalgal blooms decompose, they create hypoxic or anoxic conditions in the sediment, triggering a reduction in redox potential and an increase in toxic sulfide and ammonium concentrations [136]. This creates toxic sediment environments, as sulfide buildup inhibits root function and rhizome metabolism, often leading to die-back events, such as those observed in Florida Bay and other tropical lagoons [137]. The decomposition process also stimulates microbial activity, which accelerates the breakdown of recalcitrant organic carbon in seagrass detritus through cometabolism [22]. Laboratory experiments demonstrate that higher density macroalgal addition can stimulate approximately 20% greater decomposition of seagrass biomass and double greenhouse gas emissions, ultimately reducing blue carbon sequestration and preservation in seagrass meadows [22].

Epiphytic microalgae establish another dimension of biogeochemical interference with seagrass physiology. During nighttime, the combination of water column hypoxia and epiphytic respiration creates anoxic conditions at the leaf surface, restricting oxygen availability for plant respiration and transport to belowground tissues [138] (Fig. 4). These anoxic microenvironments enable anaerobic microbial processes which can produce harmful compounds like nitric oxide through denitrification [138]. In daylight, the photosynthetic activity within these biofilms dramatically shifts inorganic carbon equilibria, inducing elevated pH conditions and severely reducing molecular carbon dioxide availability at the leaf surface [124, 125]. Measurements show that CO_2 concentrations at the leaf surface can decrease dramatically (from $13.8 \mu\text{mol L}^{-1}$ on bare leaves to $0.09 \mu\text{mol L}^{-1}$ with epiphytic coverage), forcing seagrasses to rely on energy-intensive bicarbonate utilization [125].

Algal Toxin

Investigations across diverse coastal regions have detected potentially toxic microalgae in seagrass ecosystems. A study in southern Brazil's Paranaguá Bay Estuarine Complex identified 18 potentially toxic benthic microalgae species on various substrates including seagrass, including 7 dinoflagellates, 9 cyanobacteria, 1 diatom, and 1 euglenophyte. The abundance of these toxic microalgae varied seasonally, with dinoflagellates dominating in certain months [139]. Studies at Pegametan Bay identified 27 plankton species in seagrass habitats, including the dinoflagellate *Dinophysis* sp., with 6 diatom and 2 non-litoral species reaching bloom concentrations exceeding 10^4 cells·L⁻¹ [59].

Toxin-producing dinoflagellates such as *P. lima* and *K. brevis* release harmful compounds which disrupt seagrass ecosystems by interfering with nutrient cycling and trophic cascade effects [23, 140]. Seagrass communities serve as substrates for toxin-producing microalgae, creating direct pathways for toxin accumulation and transfer [141]. The benthic dinoflagellate *P. lima*, often found on seagrass, produces okadaic acid (OA) and dinophysistoxins (DTX), which are the causative phycotoxins responsible for diarrhetic shellfish poisoning (DSP) [142]. Studies in Moreton Bay, Australia, detected OA on multiple seagrass species, with the highest concentrations found on *Halophila spinulosa*, creating potential exposure pathways for herbivores like dugongs and green turtles that consume seagrass [23]. The red tide caused by *K. brevis* produces brevetoxins that cause massive fish kills, marine mammal mortalities, and benthic community die-offs when organisms consume toxic filter-feeding species growing on seagrasses or are directly exposed to toxins remaining on the seagrasses [143, 144]. Research has demonstrated that fish and seagrass can accumulate high concentrations of brevetoxins, serving as toxin vectors during dolphin and manatee mortality events [145]. Notably, brevetoxins and their metabolites persist in seagrass communities long after blooms subside, providing continued sources for manatee intoxication [145].

Beyond dinoflagellates, cyanobacterial toxins also threaten seagrass ecosystems. *L. majuscula*, a benthic filamentous cyanobacterium commonly found in tropical and subtropical oceans, produces tumor-promoting substances such as lyngbyatoxin A (LA) and debromoaplysiatoxin (DAT) [146]. This cyanobacterium grows epiphytically on seagrass in regions where green turtles suffer from fibropapillomatosis (FP), a potentially fatal neoplastic disease [147]. Studies in Australia and Hawaii found *L. majuscula* contributed less than 2% of green turtle dietary intake during non-bloom conditions, suggesting low-level chronic exposure to tumor-promoting compounds [146]. Tissue from deceased green turtles tested positive for LA, and estimated dosage calculations showed a positive correlation between

LA exposure and FP lesion prevalence, suggesting these naturally produced compounds may play a role in marine turtle disease etiology [146].

Allelopathic compounds from macroalgae can also negatively impact seagrass physiology [148] (Fig. 4). Research on the invasive chlorophyte *C. racemosa* var. *cylindracea* revealed that its secondary metabolite caulerpenyne exhibited phytotoxic effects on the native seagrass *C. nodosa*, reducing photosynthetic performance [149]. This allelopathic mechanism may contribute to the competitive success of invasive macroalgae over native seagrasses.

Seagrass Defenses Against HABs

Allelopathy

Challenged with HABs events, seagrasses have employed sophisticated defense mechanisms to protect themselves. One notable defense is allelopathy, which involves the production of chemical compounds that inhibit the growth of competing algae or other organisms [150]. A comprehensive survey has documented 154 chemically defined natural products across around 70 seagrass species worldwide, with most bioactivity studies focusing on their potential cytotoxic, antimicrobial, or antimacrofouling effects [151]. As early as 1980, extracts from *Z. marina* were found to inhibit the growth of microalgae (such as diatoms and dinoflagellates) and bacteria [152]. More recent studies have demonstrated the allelopathic effects of *Zostera noltei* and *C. nodosa* on benthic algae, with *Z. noltei* inhibiting *P. lima* and *C. nodosa* inhibiting *Alexandrium pacificum* significantly, though the specific chemicals involved remain unidentified [150]. Additionally, fresh *Enhalus acoroides* leaves and crude extracts from dried *E. acoroides* inhibited both the growth and photosynthetic activity of *P. concavum* [153]. Further research on *Z. marina* and *Zostera noltii* extracts showed significant dose-dependent inhibition of *Alexandrium catenella*, likely due to phenolic compounds [154]. Similarly, methanol extracts from *Syringodium isoetifolium* and *Cymodocea serrulata* have demonstrated inhibitory effects on various microalgal strains, and lipidic metabolites were the bioactive constituents of seagrass extracts [155]. Table 3 summarizes the allelopathic effects observed in various seagrass species, including both identified and unknown compounds, and their potential biological activities. Despite these advances, there are still significant gaps in our understanding of the specific allelochemicals involved, how they are produced, released, and their ecological role in natural conditions. Besides, further research is needed to explore how these interactions vary with concentration, their persistence in marine ecosystems, and potential synergistic effects among multiple compounds.

Table 3 Allelopathic effects observed in various seagrass species

Seagrass	Treatment	Target algae	Active chemicals	Reference
<i>Zostera marina</i>	Methanol extracts of leaves which had been dead from a few days to 2 weeks	<i>Cylindrotheca fusiformis</i> <i>Nitzschia angularis</i> <i>Nitzschia frustulum</i> <i>Nitzschia longissima</i> <i>Skeletonema costatum</i> <i>Platymonas</i> sp. <i>Protogonyaulax tamarensis</i> <i>Gonyaulax polyedra</i>	Unknown	[152]
<i>Zostera noltei</i>	Fresh leaves	<i>Prorocentrum lima</i>	Unknown	[150]
<i>Cymodocea nodosa</i>		<i>Alexandrium pacificum</i>		
<i>Enhalus acoroides</i>	Fresh leaves	<i>Prorocentrum concavum</i>	Unknown	[153]
	Aqueous extracts of leaves			
<i>Zostera marina</i> L.	Methanolic and aqueous extracts of fresh and detrital leaves	<i>Alexandrium catenella</i>	Phenolic acids	[154]
<i>Zostera noltii</i> Hornemann				
<i>Syringodium isoetifolium</i>	Organic solvent (dichloromethane, acetone and methanol) extracts of dried seagrasses powder	<i>Pleurosigma elongatum</i> <i>Thalassiothrix frauenfeldii</i> <i>Nitzschia sigma</i> <i>Navicula longa</i> <i>Astreonellopsis glacialis</i>	Lipidic metabolites	[155]
<i>Cymodocea serrulata</i>				
<i>Enhalus acoroides</i>	Aqueous extract of dried seagrasses powder	<i>Phaeocystis globosa</i>	Luteolin-7-O-glucuronide	[156]

Algicidal Bacteria

Seagrass beds host diverse communities of algicidal bacteria within biofilms on their surfaces. Sakami et al. found that algicidal strains were significantly more abundant on *Z. marina* leaves and in the surrounding seawater within seagrass beds compared to external seawaters, suggesting that seagrass beds are a major source of these bacteria [157]. Biofilms on seagrass leaves provide a habitat and nutrients that support the growth of these bacteria, which in turn help control HABs [158, 159]. For example, *Z. marina* is a well-studied hotspot for algicidal bacteria, with densities reaching up to 10^8 CFU per gram of leaf weight [160]. These bacteria include strains from α -proteobacteria, β -proteobacteria, and Bacteroidetes [160, 161]. Notably, a strain from the family Flavobacteriaceae has shown strong algicidal activity, effectively inhibiting harmful algae like *Alexandrium tamarense* and other HAB species, including *Chattonella antiqua*, *H. akashiwo*, *Heterocapsa circularisquama*, and *Chaetoceros mitra* [162]. The abundance of algicidal bacteria, such as *Alteromonas* spp., varies seasonally, with higher densities observed during peak eelgrass growth in June and July and a decline in September as eelgrass withers [157, 163]. This pattern suggests that active eelgrass growth stimulates the proliferation of these bacteria. Other seagrass species, such as *Enhalus acoroides* and *Thalassia hemprichii*, also harbor endosymbiotic bacteria with algicidal properties, capable of inhibiting the growth of dinoflagellates like *Porphyridium* sp. and diatoms such as *Nitzschia* sp.. The distribution of algicidal bacteria in seagrass ecosystems and target algae is summarized in Table 4.

Algicidal bacteria employ diverse mechanisms to control phytoplankton populations, operating through either direct or indirect attack strategies. Direct attack requires direct attachment to algal cells, while indirect mechanisms involve the secretion of algicidal compounds that diffuse through the water column [159]. This diversity of interaction modes allows for complex regulation of algal populations under different environmental conditions. Seagrass beds serve as critical natural reservoirs for these beneficial bacteria, continuously releasing them into surrounding waters and thereby establishing a persistent biological control system against HABs [167]. The strategic restoration and expansion of seagrass habitats therefore represents not only a blue carbon conservation measure but also a biologically based, environmentally sustainable approach to HAB prevention that harnesses natural ecological relationships rather than requiring chemical interventions [158].

Future Challenges and Solutions

Seagrasses often referred to as “coastal canaries” serve as global biological sentinels of increasing anthropogenic pressures on coastal ecosystems [101]. Over the twentieth century, widespread losses of seagrass ecosystems have positioned them among the world’s most threatened habitats [168]. This vulnerability is further exacerbated by the increasing frequency and intensity of HABs. Projections under intermediate greenhouse gas emissions scenarios suggest that seagrass meadows could face a significant reduction of approximately 5.3% ($\pm 2.9\%$) of their current distribution

Table 4. Algicidal bacteria targeting HAB species investigated in seagrass beds

Seagrasses	Target algae	Strain	Maximum density	Reference
<i>Zostera marina</i>	<i>Heterosigma akashiwo</i>	α -Proteobacteria	1.6×10^8 CFU/g	[160]
	<i>Alexandrium catenella</i>	γ -Proteobacteria Flabobacteria Actinobacteria	7.5×10^7 CFU/g	
<i>Zostera japonica</i>	<i>Heterosigma akashiwo</i>	γ -Proteobacteria	2.8×10^8 CFU/g	
	<i>Alexandrium catenella</i>	Flabobacteria	1.8×10^8 CFU/g	
<i>Zostera marina</i>	<i>Chattonella antiqua</i>	<i>Alteromonas</i> spp.	3.5×10^2 cells/ml	[157]
<i>Zostera marina</i>	<i>Heterosigma akashiwo</i>	Unknow	1.2×10^7 CFU/g	[164]
	<i>Chattonella antiqua</i>		1.8×10^7 CFU/g	
	<i>Karenia mikimotoi</i>		6.4×10^7 CFU/g	
	<i>Heterocapsa circularisquama</i>		9.2×10^6 CFU/g	
	<i>Cochlodinium polykrikoides</i>		2.8×10^7 CFU/g	
<i>Zostera marina</i>	<i>Alexandrium catenella</i>	Flabobacteria, α -Proteobacteria, Actinobacteria, γ -Proteobacteria, Bacilli	4.7×10^6 CFU/g	[161]
<i>Zostera marina</i>	<i>Alexandrium tamarense</i>	<i>Flavobacterium</i> spp.	10^4 cells ml ⁻¹	[162]
	<i>Chattonella antiqua</i>			
	<i>Heterosigma akashiwo</i>			
	<i>Heterocapsa circularisquama</i>			
	<i>Chaetoceros mitra</i>			
<i>Zostera marina</i>	<i>Chattonella antiqua</i>	<i>Alteromonas</i>	5.1×10^7 CFU/g	[163]
<i>Enhalus acoroides</i>	<i>Porphyridium</i> sp.	Unknow	10^6 cells/ml	[165]
<i>Thalassia hemprichii</i>	<i>Nitzschia</i> sp.			
<i>Zostera marina</i>	<i>Chattonella antiqua</i>	<i>Phaeobacter</i>	9.7×10^5 cells/ml	[166]

area by the end of the century [169]. These losses will be accompanied by significant spatial shifts, with expansions in the Arctic and declines in lower latitudes, particularly in tropical regions between 30°N and 30°S [170]. To address these challenges, a multi-faceted approach combining monitoring, protection, and restoration strategies is essential.

Remote sensing technology offers a powerful tool for large-scale and continuous monitoring of HABs in seagrass ecosystems, enabling rapid response and informed decision-making. For example, mapping *L. majuscula* blooms in the clear, shallow waters of Moreton Bay, eastern Australia, using a combination of field data and remotely sensed datasets, has provided marine park authorities with critical information to manage blooms effectively [63]. Additionally, monitoring early indicators of nutrient enrichment in seagrass meadows helps coastal resource managers prevent severe ecosystem disruptions, such as seagrass die-offs and the proliferation of high-biomass algal blooms [101]. Long-term data collection initiatives, like SeagrassNet, further enhance these efforts by providing valuable insights into seagrass densities and health over time [171]. Recently, the integration of artificial intelligence has significantly enhanced monitoring system capabilities. AI-driven remote sensing has already demonstrated its effectiveness in enhancing Mediterranean seagrass monitoring and conservation [172]. Machine learning models have been developed to automate

the measurement of seagrass morphological parameters, achieving high accuracy in classification and dimensional analysis [173]. AI-driven automated monitoring systems provide conservation managers with powerful, cost-effective tools for early detection of HAB events and seagrass decline, enabling rapid responses through real-time analysis of water quality parameters, spectral signatures of algal communities, and seagrass health indicators [174].

For seagrass meadows frequently affected by disturbances, improving water quality and reducing nutrient loading are essential management strategies to support recovery. Studies have shown that lower phytoplankton biomass and improved water clarity are closely linked to the restoration of more extensive seagrass coverage [175]. Reducing nutrient loading can be achieved through various measures, including upgrading wastewater infrastructure, managing stormwater runoff, and transitioning from septic systems to centralized treatment facilities [176]. In addition to addressing external nutrient sources, addressing internal sources of nutrients from sediments may also be necessary. Dredging projects can remove nutrient-rich sediments, which helps create more favorable conditions for seagrass recovery by reducing water column nutrient concentrations and improving sediment quality [93].

Seagrass protection and restoration strategies are essential components of comprehensive management plans.

Restoration efforts require long-term monitoring of seagrass densities and water quality to assess intervention success and adapt strategies accordingly [177]. While successful restoration is challenging, some projects have demonstrated promising results by combining habitat restoration with species reintroduction. For instance, the restoration of *Z. marina* in Virginia's coastal lagoons through an efficient seed collection and delivery program has led to the recovery of nearly 3600 ha of seagrass by 2018 [178], subsequently enhancing fish and invertebrate populations [179].

However, restoration efforts face substantial challenges in terms of cost-effectiveness and time investment. With median costs of 74,594 US\$ ha⁻¹ year⁻¹ and a mean return on investment period of 71 years [180], protection of existing seagrass meadows emerges as a crucial priority. Protection measures can include establishing marine protected areas (MPAs), implementing voluntary codes of conduct, and promoting alternative low-impact livelihoods and fishing practices [177]. Community engagement plays a vital role in these conservation efforts, as public awareness of seagrass' ecological and economic value can drive investment in protection initiatives and build long-term ecosystem resilience [181]. The successful management of HABs and preservation of seagrass blue carbon resources will ultimately depend on our ability to integrate advanced monitoring technologies, implement effective protection measures, and foster strong community partnerships in the face of increasing environmental challenges.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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